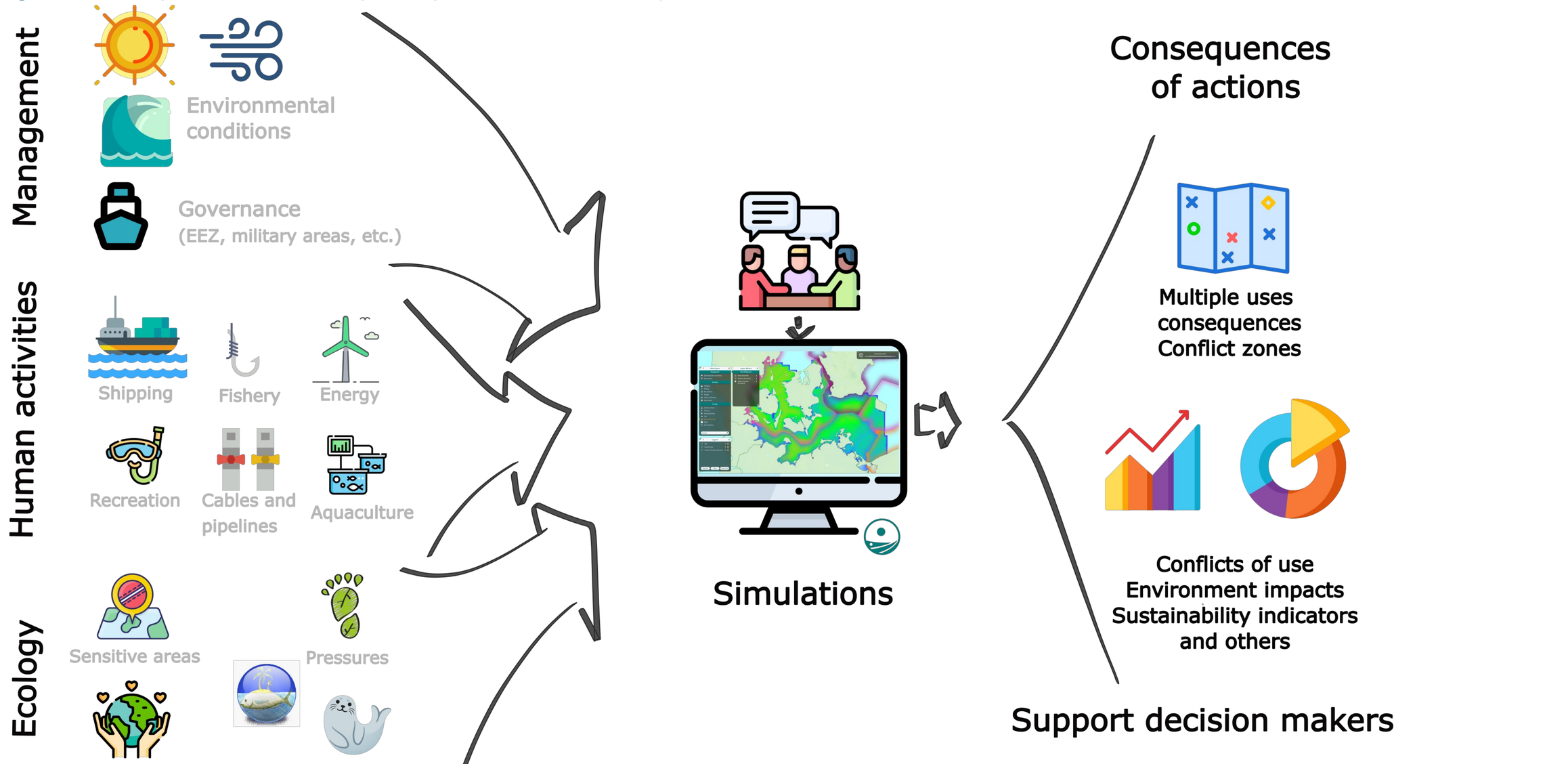
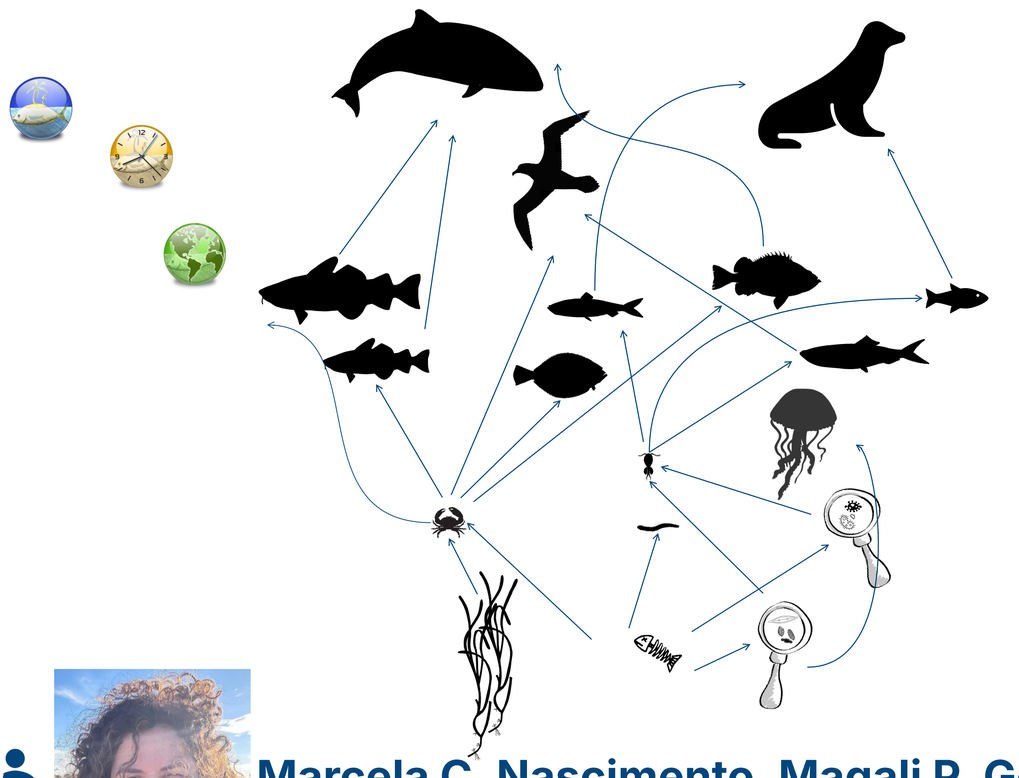
Western Baltic Sea dynamics represented in the MSP Challenge serious game

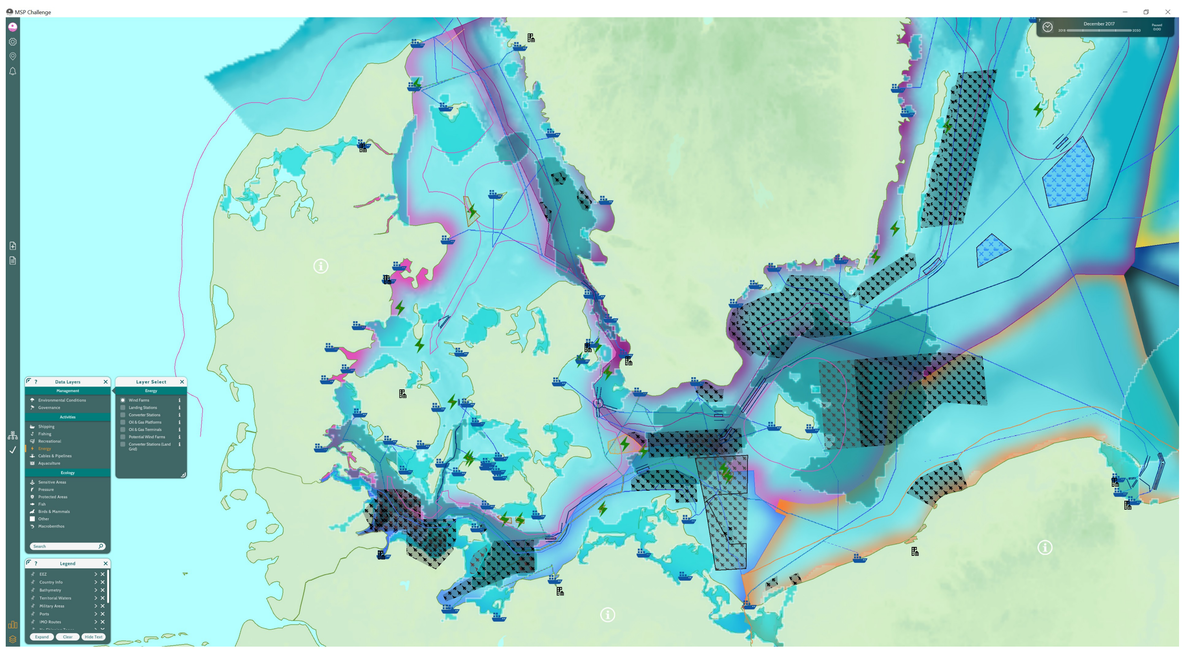
The MSP Challenge is a multi-user MSP support platform based on real geo-data of a particular sea basin. The main idea is to use a visual and interactive basis to support users in simulating Marine Spatial Planning by integrating real-life geo-spatial resources and science-based models. This platform allows users to draft spatial plans for creating new activities, altering existing activities, or seizing existing activities and simulating the consequences of such spatial plans in various ways. For more details https://www.mspchallenge.info/simulation-platform.html



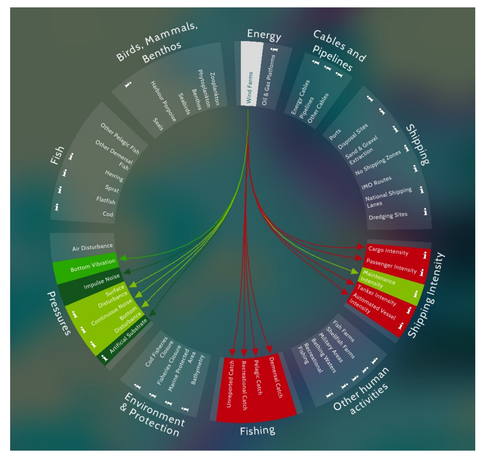
We built an MSP Challenge version for the Western Baltic Sea based on a recent Ecospace model that used as a backbone an existing Ecopath with the Ecosim model of the area (Scotti et al. 2022). The integration of the Ecospace model with multiple human uses of the sea was achieved through the development of a novel semi-quantitative method. We designed and implemented this new method to attribute the impacts of multiple, concurrent human activities over the species/functional groups. Moreover, we included new types of pressures, such as air impact, bottom vibration and two types of noise.



The Western Baltic Sea MSP Challenge version complements the existing Central Baltic Sea model and is now available for applications by external users.



The map above illustrates the western Baltic Sea MSP Challenge layout with some human activities, distribution of cod spawning areas, and Natura 2000 MPAs. On the right, the graph visualizing the groups influenced by specific human impacts and the intensity over the impacted species/functional groups





Marcela C. Nascimento, Magali P. Gonçalves, Igor Mayer and Marco Scotti

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